tion of a direction (mean of the direct and reversed series) deduced from $e_1^2 = \frac{0''.455\,\Sigma\,\Delta^2}{n-s-d+1}$, where n = number of observations, s = number of series, d = number of directions, and Δ = differences of observed and adjusted values.

The figure adjustment of the triangulation generally proceeds on the supposition of equal weights to all directions used in the adjustment, except, as in the case of the Epping base net adjustment, where special weights are introduced.* To carry weight equations from the station adjustment into the figure adjustment is not practiced or favored in the Survey on account of the increased complexity and the doubtful advantage of the proceeding. The errors brought to light by geometrical conditions, and which have been called triangle-combination errors, are of a different character from those developed by a comparison of the measures made at a station, where defects of centering at observing and observed stations, effects of large local deflections of the vertical at a station, persistent lateral refraction along a line, etc., do not show; that is to say, weights pertaining to one operation are not those proper for the next operation. Further particulars of figure adjustment with statement of formulæ may be found in "The Transcontinental Triangulation," Part I.

For the computation of the sides of the triangles Legendre's theorem was used, and since none of the lines of the oblique arc are very long the spherical excess was computed by the simple formula

$$\frac{ab \sin C}{2\rho_m \rho_n \sin 1''}$$

where C = the plane angle included by the sides a and b and the subscripts to ρ refer to the radius of curvature in the meridian and in the prime vertical. Appendix No. 9, Coast and Geodetic Survey Report for 1894, page 291, contains a table to facilitate the computation of the spherical excess, the argument being the latitude of the center of the triangle. The computation of the geodetic positions along the oblique arc is made according to the formulæ and tables in the same appendix, the differences in latitude, longitude, and azimuth for two points of known distance and azimuth being given by

$$\begin{cases} -\Delta\varphi = s\cos\alpha.\ B + s^2\sin^2\alpha.\ C + (\delta\varphi)^2\ D - h\,s^2\sin^2\alpha.\ E \\ \Delta\lambda = s\sin\alpha\sec\varphi'.\ A \\ -\Delta\alpha = \Delta\lambda\sin\tfrac{1}{2}(\varphi+\varphi')\sec\tfrac{1}{2}(\Delta\varphi) + (\Delta\lambda)^3\ F \end{cases}$$

where

$$\begin{array}{l|l} \varphi' = \varphi + \Delta\varphi & \text{and} -\delta\varphi = s\cos\alpha.\ B + s^2\sin^2\alpha.\ C - hs^2\sin^2\alpha.\ E \\ \lambda' = \lambda + \Delta\lambda & \text{also } h = s\cos\alpha.\ B. \\ \alpha' = \alpha + \Delta\alpha + 180^\circ & \end{array}$$

The factors A, B, C, D, E, and F are tabulated and refer to Clarke's spheroid of 1866. Their logarithmic values are given for every minute of angle between latitudes of 18° and 72°.

A description of each station used in the base nets is given. The description of the other stations of the triangulation can be obtained upon application to the Coast and Geodetic Survey, Washington, District of Columbia.

*When special weights are introduced for the several directions at a station, they are deduced from the rough expression of the square of the probable error given by $e_1^2 = \frac{0''.455\,\Sigma\,\Delta^2}{s\,[xx]}$ where [xx] represents the diagonal coefficient of the direction in the normal equation.